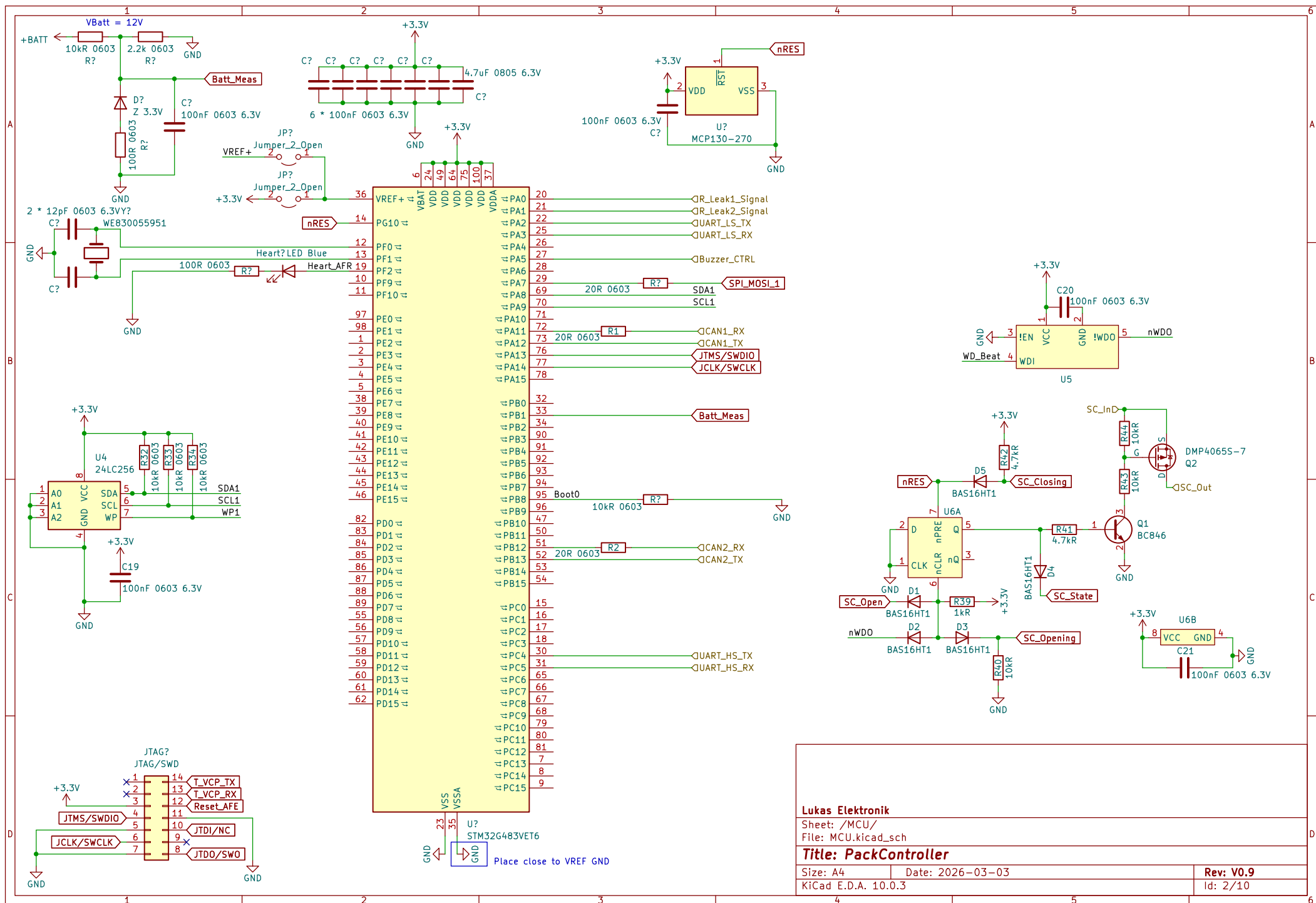




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Sheet: /MCU/

File: MCU.kicad_sch

Title: PackController

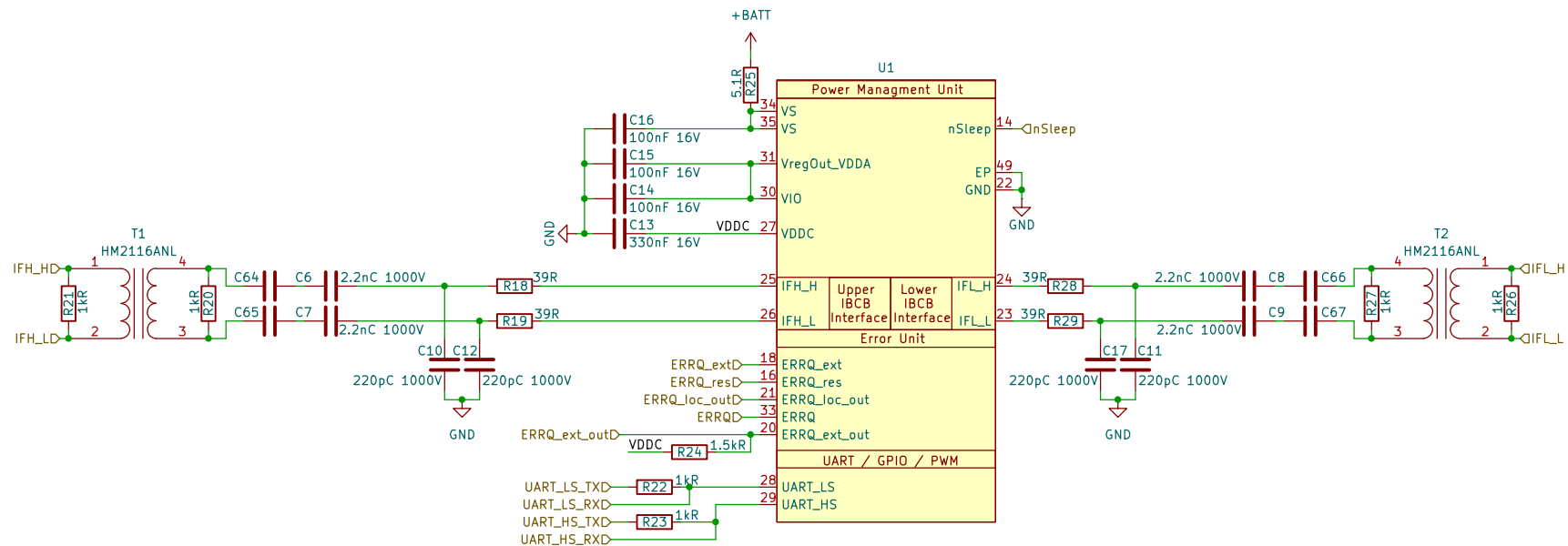
Size: A4

Date: 2026-03-03

KiCad E.D.A. 10.0.3

Rev: V0.9

Id: 2/10



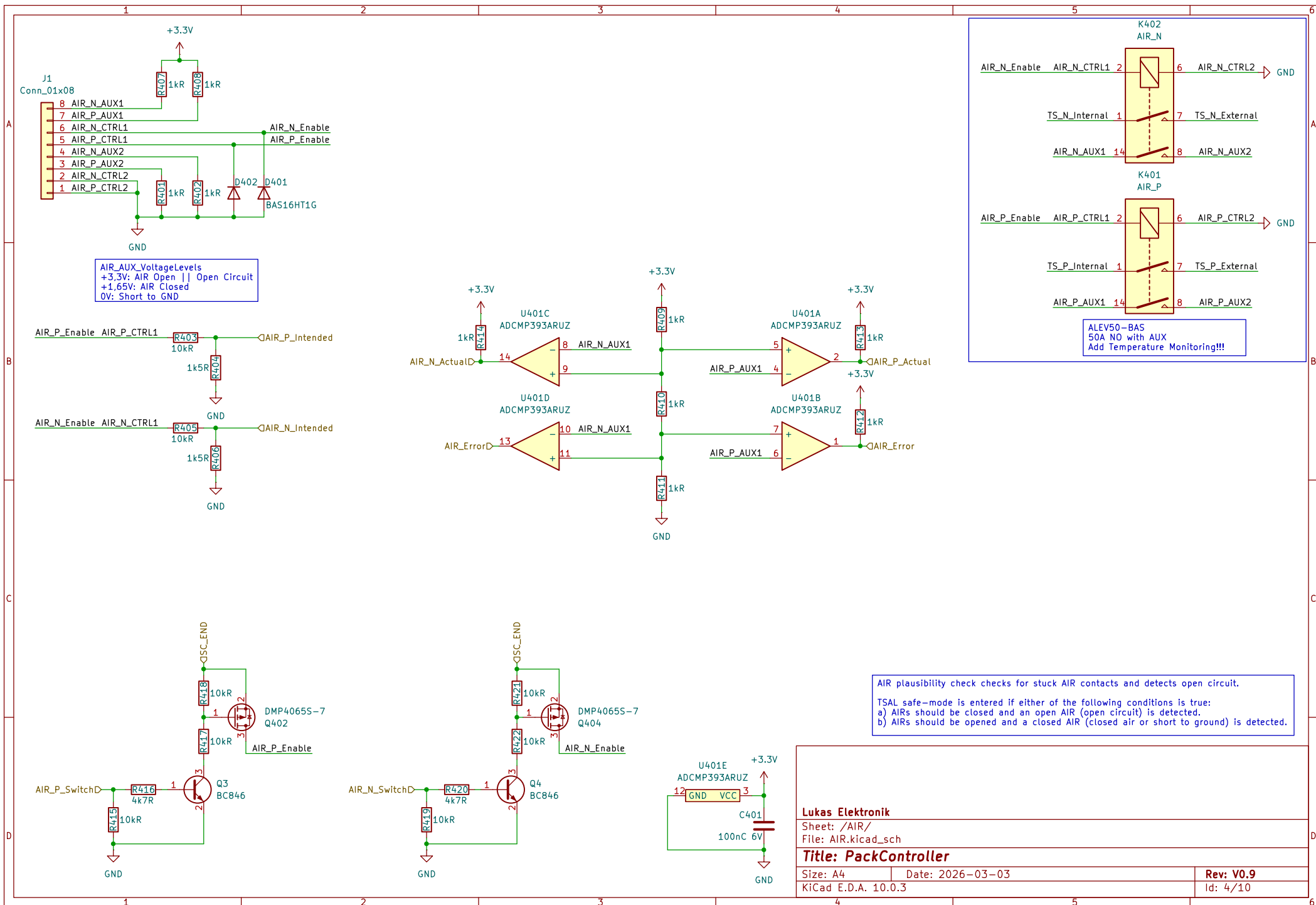
Lukas Elektronik

Sheet: /StackCommInterface/
File: StackCommInterface.kicad_sch

Title: PackController

Size: A4 Date: 2026-03-03
KiCad E.D.A. 10.0.3

Rev: V0.9
Id: 3/10



Lukas Elektronik

Sheet: /AIR/
File: AIR.kicad_sch

Title: PackController

Size: A4 Date: 2026-03-03

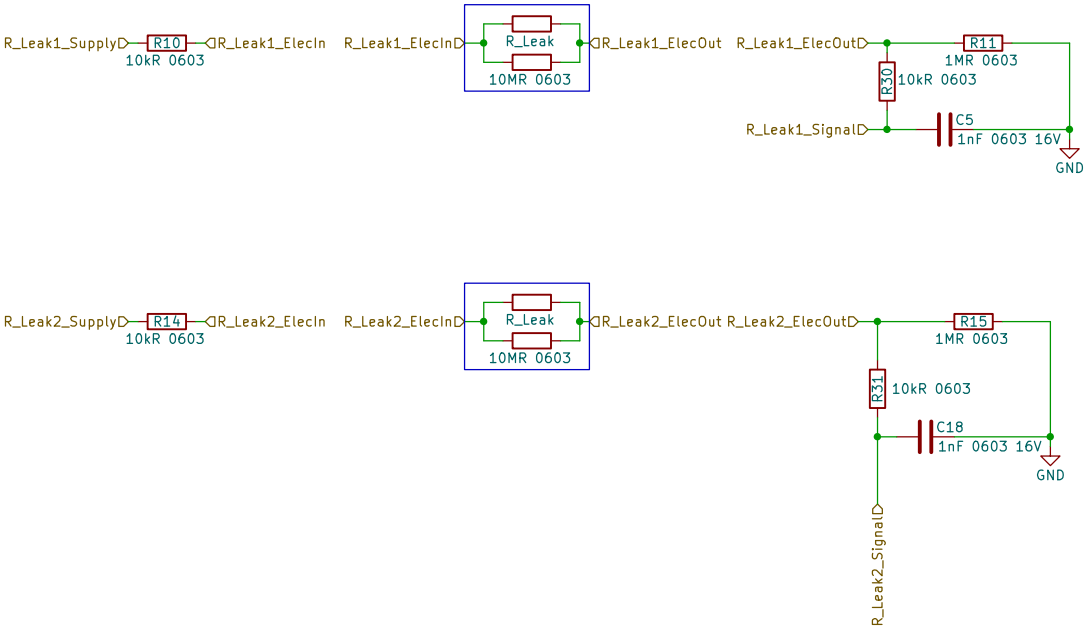
KiCad E.D.A. 10.0.3

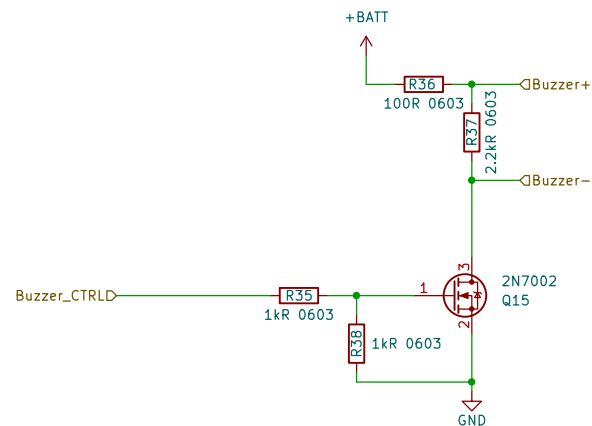
Rev: V0.9

Id: 4/10

Id: 5/10

Leakage sensor:
Flex electrodes form R_LEAK. R_EOL = 10M is mounted on the flex sensor, parallel to the electrodes, for open-wire detection.
 $R_SENSOR = R_LEAK \parallel R_EOL$
Measurement:
Pulse R_Leak_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/Hi-Z. Suggested scan rate: 1Hz.
ADC conversion, assuming ADC reference = VDD and R_Leak_Supply = VDD:
 $x = ADC_raw / ADC_max$
 $R_SENSOR = R_REF * (1/x - 1) - R_SER_SUPPLY$
Nominal:
R_SER_SUPPLY = 10k
R_REF = 1M
R_ADC_SER = 10k
C_ADC = 1nF
R_EOL = 10M on flex
Expected:
disconnected/open: ADC $\approx$ 0
dry connected: R_SENSOR $\approx$ 10M
wet/leakage: R_SENSOR decreases
Initial thresholds, validate with real coolant:
warning: R_SENSOR < 3M
leak: R_SENSOR < 1M
severe leak: R_SENSOR < 300k





Passive piezo buzzer driver.
Target part: CPT-2305-90PM, externally driven, 12 Vp-p, 4 kHz, typ. 24 nF.

Drive with STM32 PWM:
 $f_{\text{PWM}} \approx 4 \text{ kHz}$
duty $\approx 50 \%$

Q15 is a low-side NMOS switch.
R37 discharges the piezo during MOSFET off-time:
 $\tau = R37 * C_{\text{PIEZO}} \approx 2.2k * 24nF \approx 53\mu s$.

R36 limits peak current and EMI.
No additional series coupling capacitor required.

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Sheet: /Buzzer/

File: Buzzer.kicad_sch

Title: PackController

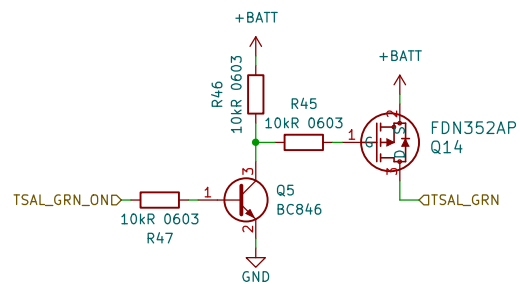
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Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 7/10



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Sheet: /TSAL_Logik/

File: TSAL_Logik.kicad_sch

Title: PackController

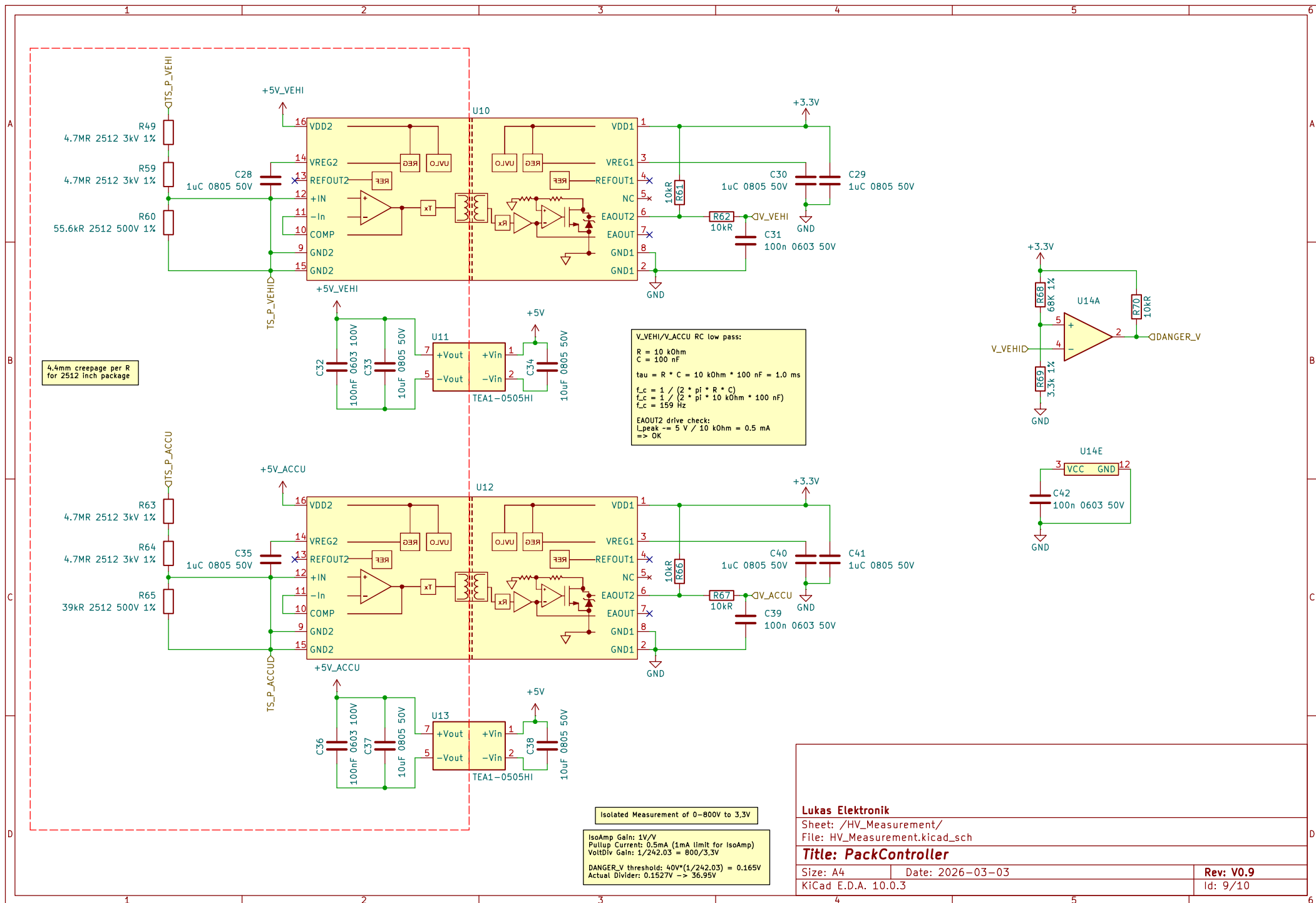
Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 8/10



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Sheet: /HV_Measurement/
File: HV_Measurement.kicad_sch

Title: PackController

Size: A4 Date: 2026-03-03
KiCad E.D.A. 10.0.3

Rev: V0.9
Id: 9/10

